Synthetic American Option Pricing via Jump-HMM-Driven Heston Implied Volatility

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Abstract

Generating realistic synthetic option prices requires implied volatility (IV) as an input, yet IV itself is derived from observed option prices—creating a circular dependency that limits synthetic data availability. In this study, we addressed this challenge by developing a three-stage pipeline that produces self-consistent American option prices: (i) a Jump Hidden Markov Model (JumpHMM) generated realistic multi-asset price paths with heavy tails, volatility clustering, and cross-asset dependence via a Student-t copula; (ii) a modified Heston stochastic variance process converted those price paths into IV, where the mean-reversion target θ was driven by the HMM regime state, days to expiration, moneyness, and an aggregate market mood indicator; and (iii) a Cox–Ross–Rubinstein binomial tree priced American options with early exercise using the pre-computed IV. A central design choice was the equilibrium initialization $v_0 = \theta(s_0, \mathrm{DTE}, K/S_0, M_0)$: each strike–expiration pair started at its own mean-reversion target, so the smile, skew, and term structure of the IV surface emerged automatically without external calibration data. We calibrated a four-parameter smile function on NVDA options and demonstrated that the smile shape generalized across AMD, MU, and INTC with only a scalar level adjustment per ticker (cross-ticker RMSE of 1–5% IV). The framework is implemented as an open-source Julia package designed to serve as a synthetic data generator for downstream machine-learning and risk-analysis applications.

1 Introduction

Pricing American-style options requires implied volatility (IV) as an input, yet IV itself is an empirical quantity extracted from observed option prices Black and Scholes [1973]. This circularity poses a practical problem for synthetic data generation: existing approaches either take IV from the market and feed it into a pricing model, or generate price paths without a consistent IV surface. The result is a scarcity of realistic synthetic option data for training machine-learning models, backtesting trading strategies, and conducting scenario-based risk analysis.

In this study, we addressed this gap by developing a framework in which IV is an *output* of a structural model of equity returns, not an input. The pipeline chained three stages:

1. **Synthetic price generation.** A Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner [2025] produced multi-asset price paths with realistic heavy tails, negligible linear autocorrelation, and persistent volatility clustering. Cross-asset dependence was imposed via a Student-t copula.
2. **Implied volatility generation.** A modified Heston stochastic variance process Heston [1993] converted those price paths into IV paths. The mean-reversion target θ was a func-

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tion of the HMM regime state, days to expiration (DTE), moneyness, and an aggregate market mood indicator.

3. **Option pricing.** A Cox–Ross–Rubinstein (CRR) binomial tree Cox et al. [1979] converted the pre-computed IV into American option prices with early exercise.

The key contributions of this study were:

- A *hybrid θ-function* that linked the Heston mean-reversion target to the JumpHMM regime state, producing regime-dependent volatility dynamics (Section 3.2.1).
- An *equilibrium initialization* $v_0 = \theta(t = 0)$ that gave each (ticker, strike, DTE) triple its own initial IV, so the smile, skew, and term structure emerged automatically without external IV data (Section 3.2.3).
- An *endogenous market mood* signal—the fraction of tickers in tail HMM states—that modulated IV without introducing circularity (Section 3.2.4).
- Calibration on real option chains demonstrating that a four-parameter smile function generalized across tickers with only a scalar level adjustment (Section 4).

The framework is implemented as an open-source Julia package and is designed to serve as a synthetic option market generator for downstream applications.

2 Background

2.1 Jump Hidden Markov Model

The Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner [2025] was developed for generating synthetic equity time series that simultaneously preserve three stylized facts of financial returns: heavy tails, negligible linear autocorrelation, and persistent volatility clustering. The model discretizes continuous excess growth rates into N states via equal-probability quantile bins of a fitted Laplace distribution. Each state k emitted observations from a location-scale Student-t distribution: $G_t \mid S_t = k \sim \mu_k + \sigma_k \cdot t_\nu$. At each timestep, with probability ϵ a Poisson(λ)-distributed jump forced the chain into tail states (bottom or top N_{tail} states), creating sustained dwell periods in extreme-volatility regimes. For multi-asset generation, cross-asset dependence was imposed via a Student-t copula that preserved each asset's marginal distribution while introducing realistic tail dependence.

2.2 Heston stochastic variance model

The Heston model Heston [1993] specifies the variance of the underlying asset as a mean-reverting square-root process:

$$dv = \kappa(\theta - v)\, dt + \sigma_v \sqrt{v}\, dW_v \tag{1}$$

where κ is the mean-reversion speed, θ is the long-run variance, and σ_v is the vol-of-vol. In the standard formulation, θ is a constant; our modification made θ time-varying and state-dependent (Section 3.2.1).

2.3 CRR binomial tree

The CRR model Cox et al. [1979] prices options on a recombining binomial lattice with up/down factors $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$ and risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$. American options are priced by backward induction with an early-exercise check at each node. The CRR tree is a natural choice when IV is provided exogenously, as it requires only a single volatility input per contract and handles early exercise without closed-form solutions.

3 Methods

3.1 Pipeline overview

The generator produced synthetic American option prices through a three-stage pipeline:

$$\underbrace{\text{JumpHMM(prices)}}_{\text{Stage 1}} \rightarrow \{S_t,\, s_t,\, \text{jumps}\} \rightarrow \underbrace{dv \text{ process}}_{\text{Stage 2}} \rightarrow \sigma_{\text{imp}} = \sqrt{v_t}$$
$$\rightarrow \underbrace{\text{CRR}(S_t,\, \sigma_{\text{imp}},\, K,\, \text{DTE},\, r_f)}_{\text{Stage 3}} \rightarrow P_{\text{american}} \tag{2}$$

Stage 1 produced correlated multi-asset price paths S_t, hidden state sequences s_t, and jump indicators via the JumpHMM. Stage 2 simulated the modified Heston variance process to obtain implied volatility. Stage 3 priced each contract via the CRR binomial tree.

3.2 Modified Heston variance process

We modified the standard Heston process (1) by making the mean-reversion target time-varying:

$$dv = \kappa\big(\theta(t) - v\big)\, dt + \sigma_v \sqrt{v}\, dW_v \tag{3}$$

3.2.1 Hybrid θ-function

The mean-reversion target decomposed into a regime-dependent base level, a market mood modulator, and a contract-specific adjustment:

$$\theta(t) = \theta_{s_t} \cdot \big(1 + \gamma \cdot M_t\big) \cdot \psi(\text{DTE}_t,\ K/S_t) \tag{4}$$

where θ_{s_t} is a learned variance level for HMM state s_t (one value per state, $s_t \in \{1, \ldots, N\}$); γ is the market mood sensitivity; $M_t \in [0, 1]$ is the aggregate market mood (Section 3.2.4); and $\psi(\text{DTE}, K/S)$ is the smile–term-structure adjustment (Section 3.2.2). The regime-dependent base θ_{s_t} linked the Heston process directly to the JumpHMM state sequence: when the HMM transitioned into a tail state (via a Poisson jump or an extreme Markov transition), θ_{s_t} increased, elevating the IV target. This mechanism produced the leverage effect—large negative returns drove the HMM into low-numbered tail states with $p_{\text{neg}} \approx 0.52$, which in turn elevated θ—as an emergent property of the return dynamics rather than as a separate model component.

3.2.2 Smile and term-structure function ψ

The function ψ captured the shape of the IV surface with four parameters:

$$\psi(\text{DTE},\ K/S) = \exp\Big(\beta_1 \ln(\text{DTE}) + \beta_2 \ln(K/S) + \beta_3 \ln(\text{DTE}) \ln(K/S) + \beta_4 \big[\ln(K/S)\big]^2\Big) \tag{5}$$

where β_1 controlled the term-structure decay (positive values produced contango—IV decreased as expiration approached); β_2 controlled the skew (negative values produced put skew, with OTM puts having higher IV than OTM calls); β_3 captured the DTE–skew interaction (the skew flattened at longer maturities); and β_4 controlled the smile curvature (positive values produced the U-shaped vol smile, with both wings elevated relative to ATM).

3.2.3 Equilibrium initialization

Rather than specifying a separate initial variance v_0, we initialized the variance process at its mean-reversion target:

$$v_0 = \theta(s_0,\ \text{DTE},\ K/S_0,\ M_0) \tag{6}$$

This ensured that each (ticker, strike, DTE) triple started at its own equilibrium implied volatility. The initial IV surface inherited the smile, skew, and term structure encoded in ψ without requiring any external IV data. Different contracts on the same underlying received different v_0 values, and the variance process immediately began mean-reverting around its local $\theta(t)$. The value of this design choice lay in eliminating v_0 as a free parameter: the model did not require a separate calibration of the initial IV surface, as it was determined by construction from θ.

3.2.4 Market mood

The market mood M_t was defined as the fraction of tickers currently occupying tail HMM states:

$$M_t = \frac{1}{N_{\text{tickers}}} \sum_{i=1}^{N_{\text{tickers}}} \mathbf{1}\Big\{ s_t^{(i)} \le N_{\text{tail}} \text{ or } s_t^{(i)} > N - N_{\text{tail}} \Big\} \tag{7}$$

This signal was fully endogenous to the JumpHMM copula model—it required no external data (e.g., VIX) and introduced no circularity. When multiple tickers simultaneously occupied tail states, the mood rose, elevating θ and thus IV across all contracts. This mechanism produced correlated IV spikes during market stress events by construction.

3.2.5 Discretization

The variance process was discretized via Euler–Maruyama with a reflecting boundary at zero:

$$v_{t+1} = \Big| v_t + \kappa\big(\theta_t - v_t\big)\Delta t + \sigma_v \sqrt{\max(v_t, 0)}\,\sqrt{\Delta t}\, Z_t \Big|, \quad Z_t \sim \mathcal{N}(0,1) \tag{8}$$

The reflecting boundary allowed θ to vary freely across HMM states without requiring the Feller condition $2\kappa\theta > \sigma_v^2$ to hold at every timestep Andersen [2008]. This was a practical necessity, as tail states could drive θ to values where the Feller condition would otherwise be violated.

4 Calibration and Validation

4.1 Data

The empirical foundation of this study was a set of option chains for four semiconductor stocks: NVDA, AMD, MU, and INTC. All chains were single-date snapshots sourced from barchart.com (October–November 2025) with DTEs ranging from 41 to 80 days. We filtered each chain to moneyness $K/S \in [0.70, 1.30]$ and IV $< 200\%$ to exclude illiquid deep ITM/OTM contracts with unreliable IV. Table 1 summarizes the dataset after filtering.

Table 1: Option chain summary. Contract counts are reported after filtering to the tradeable moneyness range.

Ticker	Spot ($)	DTE	ATM IV (%)	Contracts
NVDA	207.04	80	44.6	142
AMD	225.78	58	59.0	36
MU	237.02	41	72.4	34
INTC	37.78	65	53.6	46

4.2 Calibration procedure

Since each chain was a single-date snapshot, all observations shared one HMM state s_0 and market mood M_0. The model IV at a given strike reduced to:

$$\sigma_{\text{model}}(K) = \sqrt{\theta_{\text{base}} \cdot \psi(\text{DTE},\ K/S)} \tag{9}$$

where $\theta_{\text{base}} = \theta_{s_0} \cdot (1 + \gamma M_0)$ absorbed the state and mood into a single scalar. We fixed $\beta_1 = 0.20$ (term-structure slope) because single-DTE snapshots cannot identify it; multi-expiration data from the same observation date would allow direct calibration of β_1 (Section 6). The remaining parameters $(\theta_{\text{base}}, \beta_2, \beta_3, \beta_4)$ were fitted by minimizing the mean squared IV error on NVDA:

$$\min_{\theta_{\text{base}},\beta_2,\beta_3,\beta_4} \frac{1}{n} \sum_{i=1}^{n} \big(\sigma_{\text{model}}(K_i) - \text{IV}_{\text{market},i}\big)^2 \tag{10}$$

using Nelder–Mead optimization. We chose NVDA as the calibration ticker because it had the richest chain (142 contracts after filtering).

4.3 Results

The calibrated parameters are reported in Table 2. The negative β_2 produced put skew; the positive β_4 produced smile curvature; and the interaction term β_3 captured how the skew flattened at longer maturities. The calibration achieved an RMSE of 1.16% IV on NVDA.

Table 2: Calibrated smile parameters (fitted on NVDA, DTE = 80).

Parameter	Interpretation	Value
β_1	Term-structure decay (fixed)	0.20
β_2	Skew (asymmetry)	-2.07
β_3	DTE $\times$ skew interaction	0.34
β_4	Smile curvature	1.43
θ_{base}	Variance level (NVDA)	0.084

4.4 Cross-ticker validation

To determine whether the smile *shape* generalized across tickers, we applied the $(\beta_1, \beta_2, \beta_3, \beta_4)$ calibrated on NVDA to AMD, MU, and INTC, re-optimizing only the scalar θ_{base} per ticker. Table 3 reports the root-mean-square IV error; Figure 1 shows the NVDA calibration fit; and Figure 2 compares the model smile to market data across all four tickers by moneyness.

Table 3: Cross-ticker RMSE (% IV). The smile shape $(\beta_2, \beta_3, \beta_4)$ was shared from NVDA; only θ_{base} was re-fitted per ticker.

Ticker	θ_{base} (IV level)	RMSE (% IV)
NVDA (calibration)	28.9%	1.16
AMD	38.7%	2.34
MU	49.0%	3.99
INTC	35.9%	4.65

The NVDA fit confirmed that four parameters sufficed to capture the smile shape within the tradeable moneyness range (RMSE = 1.16% IV). The cross-ticker validation revealed that the shared smile shape transferred well to AMD (2.34%) and reasonably to MU (3.99%) and INTC (4.65%). The higher RMSE for MU and INTC reflected differences in skew steepness across these names—MU exhibited a steeper put skew than NVDA, while INTC had a flatter OTM call wing—suggesting that per-ticker β calibration would further reduce errors when multi-strike data is available. The residual analysis (Figure 3) confirmed that the errors were concentrated in the wings ($K/S < 0.85$ and $K/S > 1.15$), with near-ATM residuals below 1% IV for all tickers.

5 Scenario Generation

To demonstrate the framework as a synthetic data generator, we simulated 500 forward paths for NVDA over 80 trading days using the calibrated parameters from Section 4 and a JumpHMM marginal model with 50 states and tuned jump parameters. The tail states were assigned elevated θ values ($2\times$ baseline for bearish tail states, $1.5\times$ for bullish tail states), and the mood sensitivity was set to $\gamma = 0.5$.

5.1 Forward IV dynamics

Figure 4 shows the evolution of implied volatility across the 500 scenarios. The mean IV started near the calibrated ATM level and exhibited gentle term-structure decay as the contract approached expiration—consistent with the positive $\beta_1 = 0.20$ (contango). Individual paths showed stochastic fluctuations driven by the vol-of-vol parameter σ_v, with occasional spikes when the HMM transitioned into tail states. The 5th–95th percentile envelope widened over time, reflecting the accumulation of variance process noise. The market ATM IV of 44.6% sat within the model's distributional

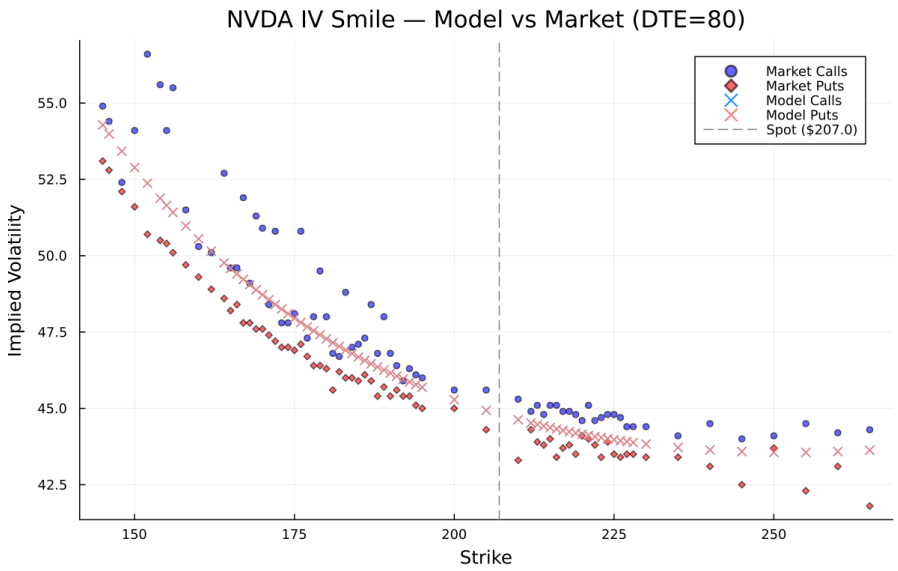

Figure 1: NVDA implied volatility smile: model (crosses) vs. market (circles) for calls and puts. The dashed vertical line marks the spot price ($207). DTE = 80.

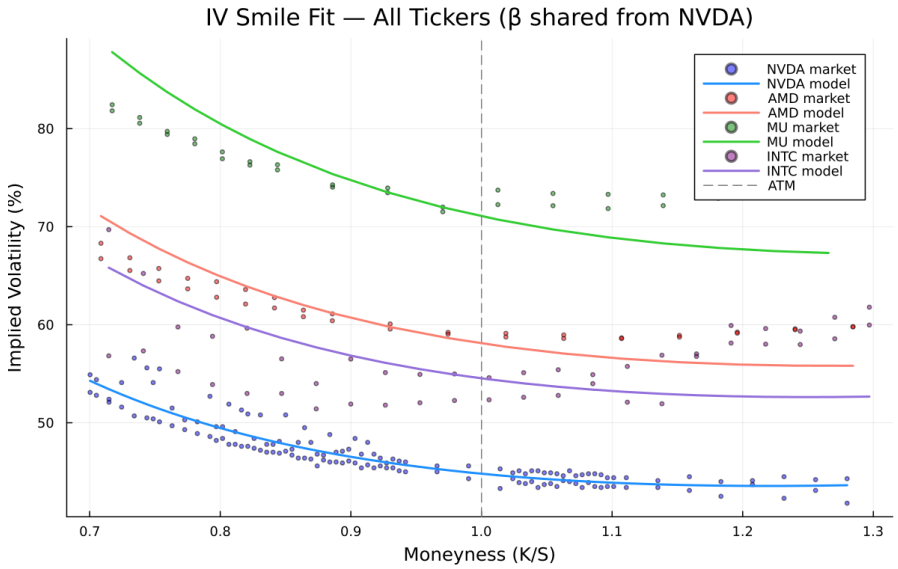

Figure 2: IV smile fit across all four tickers as a function of moneyness (K/S). The smile shape $(\beta_2, \beta_3, \beta_4)$ was shared from NVDA; only the level θ_{base} varied per ticker. Lines show model predictions; markers show market data.

range, confirming that the calibrated parameters produced IV dynamics centered on the observed level.

5.2 Option price distributions

Table 4 reports the distribution of American option prices across scenarios, evaluated at day 5 of the forward simulation. The price distributions exhibited realistic properties: OTM puts (strike $= 0.90S$) had a mean price of $12.59 with a 90% confidence interval of [$5.6, $21.1]; ATM calls were approximately $2\times$ more expensive than 10%-OTM calls; and the confidence intervals widened for contracts farther from the money. These properties are consistent with the behavior of real option markets.

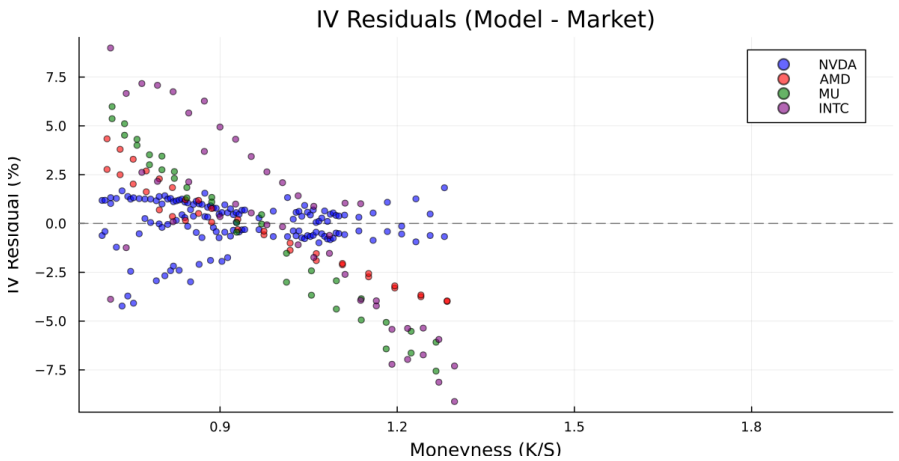

Figure 3: IV residuals (model − market) by moneyness across all four tickers. Residuals were concentrated in the wings; near-ATM errors were below 1% IV.

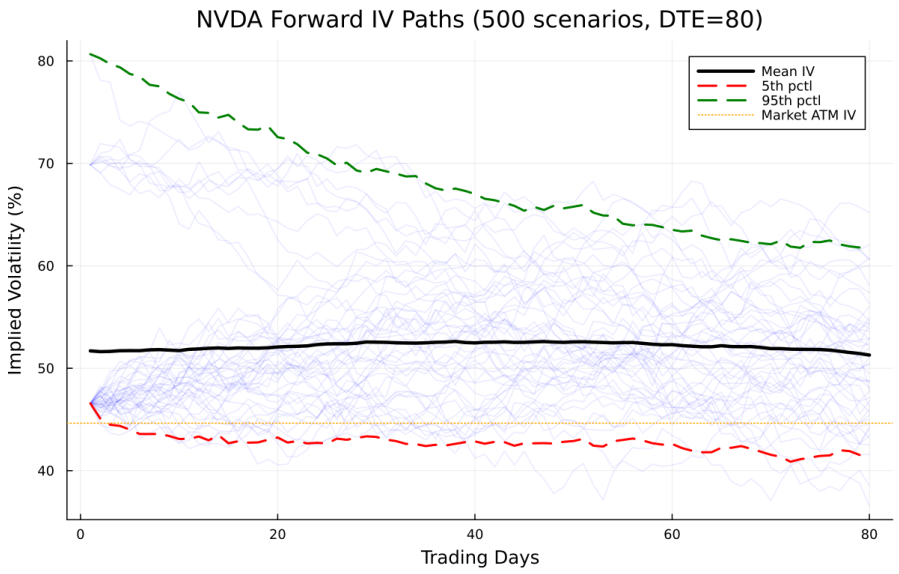

Figure 4: Forward IV paths for NVDA (500 scenarios, DTE = 80). Black line: mean IV; dashed red/green lines: 5th/95th percentiles; orange dotted line: market ATM IV (44.6%). Individual paths (blue, $\alpha = 0.08$) show stochastic fluctuations and occasional tail-state-driven spikes.

5.3 Tail event analysis

The market mood mechanism produced a natural partition of scenarios into calm ($M_t = 0$) and stressed ($M_t > 0$) paths. In multi-asset simulations with five tickers, stressed scenarios—where multiple tickers simultaneously occupied tail HMM states—produced systematically higher option prices. OTM puts were $1.83\times$ more expensive in stressed vs. calm scenarios (strike $= 0.95S$: \$6.42 vs. \$3.51), while deep OTM weekly puts showed a $4.33\times$ stress premium. This asymmetry—puts repricing more sharply than calls under stress—is consistent with the crash-protection premium observed in equity options markets and emerged naturally from the JumpHMM's negative jump bias ($p_{\text{neg}} = 0.52$) propagating through the θ-function.

Table 4: American option price distributions from 500 forward scenarios (NVDA, evaluated at $t = 5$). Strike levels are expressed relative to the spot price $S = \$207$.

Type	Strike	DTE	Mean ($)	Std ($)	90% CI ($)
Put	186 ($0.90S$)	80	12.59	4.99	[5.6, 21.1]
Put	197 ($0.95S$)	80	16.69	5.56	[8.5, 26.2]
Call	207 (ATM)	80	25.59	6.64	[15.5, 37.0]
Call	217 ($1.05S$)	80	20.94	6.29	[11.4, 31.7]
Call	228 ($1.10S$)	80	16.72	5.81	[8.1, 26.6]

6 Discussion and Future Work

The calibration and scenario results validated the core claim of this study: a modified Heston process with a state-dependent θ-function can produce realistic IV surfaces from structural return dynamics alone. The four-parameter smile function captured the essential features of the NVDA IV smile (RMSE = 1.16% IV), and the smile shape generalized across semiconductor names with only a scalar level adjustment, at the cost of higher wing residuals for tickers with steeper or flatter skew than NVDA. The equilibrium initialization $v_0 = \theta(t = 0)$ eliminated v_0 as a free parameter and produced the smile, skew, and term structure by construction.

6.1 Limitations

Term-structure calibration. The parameter β_1 (term-structure decay) was fixed at 0.20 because the available option chains contained only a single expiration per ticker. Multi-expiration data from the same observation date would allow direct calibration of β_1 and its interaction with the other smile parameters; whether the contango assumption ($\beta_1 > 0$) holds uniformly across market regimes remains an open question.

Cross-ticker smile shape. The shared $(\beta_2, \beta_3, \beta_4)$ from NVDA did not perfectly transfer to all tickers—RMSE increased from 1.2% (NVDA) to 4.7% (INTC). The residual analysis (Figure 3) revealed systematic patterns in the wings, suggesting that per-ticker smile calibration would be necessary for applications requiring sub-2% IV accuracy across all names.

Static calibration. The current calibration used single-date snapshots. Time-series calibration—fitting θ_{s_t} values across multiple dates with known HMM states—would produce a more robust mapping from regime states to variance levels and would allow the mood sensitivity γ to be identified from data rather than set heuristically.

6.2 Future work

Multi-expiration calibration. With option chains at 2–3 expirations per ticker on the same date, β_1 can be identified from data and the interaction between term structure and smile shape can be characterized.

Downstream ML applications. The primary intended use of this framework is generating realistic synthetic training data for machine-learning models in finance. Concrete applications include: training option price predictors, learning delta-hedging policies via reinforcement learning, and augmenting scarce real option data for transfer learning.

Greeks and risk sensitivities. The framework naturally supports computation of Greeks (Δ, Γ, $\mathcal{V}$, Θ) via finite differences on the CRR tree. Including these in the synthetic dataset would support hedging applications.

Cross-asset IV correlation. The current mood mechanism created correlated IV dynamics through a shared stress signal. A richer model could introduce ticker-pair IV correlations through the copula structure, producing a full cross-asset IV surface that respects observed correlation patterns between individual-name and index volatility.

7 Conclusion

In this study, we developed a framework for generating synthetic American option prices with self-consistent implied volatility dynamics. The framework linked a Jump Hidden Markov Model to a modified Heston stochastic variance process via a state-dependent mean-reversion target $\theta(t)$, and priced the resulting IV through a CRR binomial tree with early exercise. The equilibrium initialization $v_0 = \theta(t{=}0)$ eliminated the initial IV surface as a free parameter, and the endogenous market mood signal produced correlated IV spikes during stress events without external data.

We calibrated a four-parameter smile function on NVDA options and demonstrated that the smile shape generalized across AMD, MU, and INTC with cross-ticker RMSE of 1–5% IV using only a scalar level adjustment per ticker. Forward scenario analysis confirmed that the generated IV dynamics exhibited realistic term-structure decay, stochastic fluctuations, and a crash-protection premium in OTM puts under stress (1.8–$4.3\times$ calm-to-stress price ratios).

The framework is implemented as an open-source Julia package and is available at https://github.com/varnerlab/heston-implied-volatility-model.

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